

Here let's mimic the proof of [1, Lemma 7.2].

By Lagrange multiplier, we see that

$$\begin{aligned} 2 \left(\int_{\mathbb{R}} |f(x)|^{2p} dx \right)^{-(p-1)/p} \int_{\mathbb{R}} f^{2p-1}(x) g(x) dx - \frac{1}{2\pi} \int_{\mathbb{R}} |\xi|^a \mathcal{F}g(\xi) \overline{\mathcal{F}f(\xi)} d\xi \\ = \frac{\lambda}{\pi} \int_{\mathbb{R}} \overline{\mathcal{F}f(\xi)} \mathcal{F}g(\xi) d\xi. \end{aligned}$$

We can remove the conjugate sign in the above equality because f is an even function. By Plancherel theorem, we see that

$$\begin{aligned} \left(\int_{\mathbb{R}} |f(x)|^{2p} dx \right)^{-(p-1)/p} \int_{\mathbb{R}} \mathcal{F}[f^{2p-1}](\xi) \mathcal{F}g(\xi) d\xi - \frac{1}{2} \int_{\mathbb{R}} |\xi|^a \mathcal{F}g(\xi) \mathcal{F}f(\xi) d\xi \\ = \lambda \int_{\mathbb{R}} \mathcal{F}f(\xi) \mathcal{F}g(\xi) d\xi. \end{aligned}$$

By writing the above equality in the following way:

$$\begin{aligned} \left(\int_{\mathbb{R}} |f(x)|^{2p} dx \right)^{-(p-1)/p} \int_{\mathbb{R}} \frac{\mathcal{F}[f^{2p-1}](\xi)}{|\xi|^{a/2}} \mathcal{F}g(\xi) |\xi|^{a/2} d\xi - \frac{1}{2} \int_{\mathbb{R}} \mathcal{F}f(\xi) |\xi|^{a/2} \mathcal{F}g(\xi) |\xi|^{a/2} d\xi \\ = \lambda \int_{\mathbb{R}} \frac{\mathcal{F}f(\xi)}{|\xi|^{a/2}} \mathcal{F}g(\xi) |\xi|^{a/2} d\xi, \end{aligned}$$

we see that for all $\xi \in \mathbb{R}$,

$$\left(\int_{\mathbb{R}} |f(x)|^{2p} dx \right)^{-(p-1)/p} \frac{\mathcal{F}[f^{2p-1}](\xi)}{|\xi|^{a/2}} - \frac{1}{2} \mathcal{F}f(\xi) |\xi|^{a/2} = \lambda \frac{\mathcal{F}f(\xi)}{|\xi|^{a/2}}.$$

Multiply $|\xi|^{a/2}$ on both sides to see that

$$\left(\int_{\mathbb{R}} |f(x)|^{2p} dx \right)^{-(p-1)/p} \mathcal{F}[f^{2p-1}](\xi) - \frac{1}{2} \mathcal{F}f(\xi) |\xi|^a = \lambda \mathcal{F}f(\xi). \quad (7.9)$$

Now multiply $(2\pi)^{-1} \mathcal{F}f(\xi)$ on both sides of (7.9) and integrate to see that

$$\left(\int_{\mathbb{R}} |f(x)|^{2p} dx \right)^{1/p} - \frac{1}{4\pi} \int_{\mathbb{R}} |\xi|^a |\mathcal{F}f(\xi)|^2 d\xi = \lambda,$$

where we have used the fact that $\|f\|_{L^2(\mathbb{R})} = 1$.

Now multiply $(2\pi)^{-1} \mathcal{F}f(\xi) |\xi|^{a/2}$ on both sides of (7.9) and integrate to see that... **The difficulties come from here. I cannot find the counter part of the integration by parts for fractional derivatives.**

References

- [1] Chen, X. and Li, W. (2004). Large and moderate deviations for intersection local times. *Probab. Theory Related Fields* **128** 213–254.